

Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

Machine learning researcher and engineer with seven years of experience developing methods and software in research and industry. MSc in Engineering Physics, specialising in Statistical Learning and Data Analytics. Research experience in representation learning, probabilistic modelling, and numerical optimisation, complemented by practical work on neural networks, embedding-based retrieval, and large-scale ML workflows. Looking to return to research, with a particular interest in generative AI.

Education

- 2015–2020 **KTH, Royal Institute of Technology – MSc. in Engineering Physics**, GPA: 5.0 / 5.0
Specialisation: Applied Mathematics, Statistical Learning and Data Analytics.
Relevant coursework: Statistical Machine Learning, Reinforcement Learning, Optimisation.
Thesis: [Probabilistic Dose Prediction for Radiation Therapy Planning](#)
- 2019 **ETH Zürich, Swiss Federal Institute of Technology – Applied Mathematics**, Exchange
- 2017–2022 **SU, Stockholm University – BSc. in Economics**, Additional Bachelor's Degree
- 2017 **ZJU, Zhejiang University**, Research exchange, Laser trimming of microring resonators.

Selected Experience

Researcher within Data Science @ RaySearch Laboratories

- Developed autoencoder-based patient embeddings and learned similarity from segmented 3D CT scans.
- Predicted dose-volume histograms and spatial dose distributions using sparse Gaussian processes.
- Redesigned optimisation algorithms to reflect clinical decision hierarchies, validated with clinicians.

Keywords: deep learning, representation learning, probabilistic modelling, Gaussian processes, numerical optimisation, clinical collaboration

[Read more...](#)

Data Scientist / ML Engineer @ Signum Life Science

- Develop automated ML workflows for pharmaceutical forecasting, from data validation and training to experiment tracking, batch inference, and reproducible evaluation.
- Design maintainable ML architecture supporting thousands of independently trained models.
- Develop embedding-based retrieval and matching workflows for domain-specific text, evaluating semantic similarity and analysing retrieval failures.

Keywords: Python, machine learning, data pipelines, model evaluation, Databricks, MLflow, Docker, Linux

[Read more...](#)

Personal Project: *iris*

- Developed a computer vision based product retrieval system combining object detection with YOLOv8, OpenCLIP embeddings, and vector similarity search.
- Built the complete application, including model fine-tuning, backend APIs and a browser extension.

Keywords: computer vision, representation learning, YOLOv8, OpenCLIP, information retrieval, vector similarity search

[Read more...](#)

Skills and Other

Technology	Python, PyTorch, TensorFlow, SQL, C++, Git, Linux, Docker, MLflow, Databricks
Methods	Representation learning, deep learning, probabilistic modelling, model evaluation
Language	Native English and Swedish, intermediate Danish

Detailed Work Experience

2024– Present **Data Scientist / ML Engineer, Signum Life Science, Copenhagen**

At Signum, I work across ML, software architecture, and project management, translating loosely defined problems in pharmaceutical analytics into maintainable production systems.

I develop embedding-based retrieval and matching workflows for domain-specific text. This involves evaluating semantic similarity, analysing retrieval failures, and improving matching reliability.

I also lead architecture and development of a forecasting platform for pharmaceutical prices, demand, and patient populations. The platform combines data validation, model training, experiment tracking, and batch inference in Databricks, supporting thousands of independently trained models. My work includes reproducible evaluation, maintainable software design, and coordination with technical and domain stakeholders.

2022–2024 **Principal Data Scientist, Valcon, Copenhagen**

At Valcon, I led ML projects across energy, IoT, pharmaceutical, and manufacturing domains, taking projects from problem formulation and model development through production deployment. I also mentored junior colleagues and delivered training in explainable AI, Git, and software architecture.

For one project, I designed and deployed an end-to-end pipeline for a real-time neural network. Using transfer learning, the model combined natural-language and high-dimensional categorical inputs to predict high-dimensional categorical outputs. I was responsible for the training pipeline, modular model architecture, evaluation, and production integration.

I received the Valcon People's Choice Award for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist, Valtech, Copenhagen**

At Valtech, I developed ML solutions for e-commerce, including time-series forecasting and customer-behaviour modelling. I built forecasting pipelines using Prophet and applied clustering to behavioural data to support personalisation and customer segmentation.

I worked with technical and commercial stakeholders to translate modelling results into operational recommendations and presented findings to senior decision-makers.

2019–2021 **Researcher within Data Science, RaySearch Laboratories, Stockholm**

At RaySearch Laboratories, I researched ML and optimisation methods for automated radiotherapy planning. The work combined deep learning, probabilistic modelling, and multi-objective optimisation in a safety-critical clinical setting.

I developed a pipeline that learned latent anatomical representations from segmented 3D CT scans using autoencoders. These representations were used to identify similar patients through a learned similarity metric and to predict dose-volume histograms and spatial dose distributions using sparse Gaussian processes.

I also redesigned lexicographic optimisation algorithms to better represent clinical decision hierarchies. I collaborated with medical physicists and clinicians to define requirements, evaluate model behaviour, and ensure that the resulting methods aligned with oncological standards and practical workflows.

Personal Projects

Project iris Developed a visual product retrieval system that detects products within webshop images and matches them to catalogue items. Combined YOLOv8 object detection with OpenCLIP embeddings and Qdrant vector similarity search. Built the supporting data collection pipeline, backend APIs, and Chromium extension to display product links within images. [GitHub repository](#).